B.TECH CSE | FULL STACK PROJECT

Smart Agriculture & Crop Management System

A centralized web platform to monitor fields, plan crops, track activities, and improve farm yields efficiently.

DEVELOPER

[Student Name] ([Roll Number])

DEPARTMENT

Computer Science & Engineering

INSTITUTION

[College Name]

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CONTEXT & OVERVIEW

Replacing Farming Guesswork with One Smart Platform

Traditional Challenges

- **Guesswork:** Irrigation and fertilizer decided by habit
- **Weather Risk:** No early warning for rain or heat
- **Manual Records:** Paper notes for crops and expenses
- **No Insights:** Little data to improve yield

Proposed Digital Platform

+ **Unified Hub:** One app for farms, crops and tasks
+ **Smart Monitoring:** Track soil, moisture and weather data
+ **Crop Advisory:** Season-wise crop and care guidance
+ **Anywhere Access:** Available 24/7 across devices

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TECHNICAL FOUNDATION

MERN Stack Architecture & Workflow

1

Frontend

React.js, Vite, HTML5, CSS3, JavaScript

2

Backend

Node.js, Express.js Framework

3

Database

MongoDB, Mongoose ODM

4

Tools

VS Code, REST API, Git, Weather API

DATA FLOW

React Frontend → REST API (JSON) → Express Controller → Mongoose Model → MongoDB

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PROCESS MAP

End-to-End Crop Management Workflow

1

Register Farm

Add farm details, land size and soil type.

2

Plan Crop

Select crop and season using advisory tips.

3

Monitor Field

Log moisture, weather and growth data.

4

Take Action

Schedule irrigation, fertilizer and pest control.

5

Harvest & Record

Record yield and update status to Harvested.

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DATA ARCHITECTURE

MongoDB Collections Design

User Collection

Primary Model

_id (PK)

Name & Email

Password (Hashed)

Role (Farmer / Admin)

Crop Collection

Core Data

_id (PK), UserID (FK)

CropName & Season

SoilType, Area

SowingDate, Status

Activity Collection

Transaction

_id (PK)

CropID (FK)

ActivityType & Date

SensorReading & Notes

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PRODUCT VISION

Planned Future Enhancements

IoT Sensor Integration

Live soil moisture and temperature feeds.

AI Crop Recommendation

Suggest best crops from soil and weather.

Disease Detection

Identify plant diseases from leaf photos.

Mobile Application

Dedicated iOS and Android versions.

Yield Analytics

Dashboards on yield, cost and profit trends.

Real-Time Alerts

SMS and push alerts for weather and irrigation.

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CONCLUSION

Thank You!

Smart Agriculture & Crop Management System provides a streamlined, data-driven way to grow better crops and manage farms efficiently.

Questions & Answers

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